RAJALAKSHMI ENGINEERING COLLEGE

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M2

Started on	Thursday, 24 September 2026, 10:36 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:49 AM
Time taken	13 mins 52 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 10.00 out of 10.00

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*

Output:*6*

Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1

3+3

1+1+1+3

1+1+3+1

1+3+1+1

3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n;
    scanf("%d", &n);

    long long *dp = (long long *)calloc(n + 1, sizeof(long long));
    dp[0] = 1;

    for (int i = 1; i <= n; i++) {
        dp[i] = dp[i - 1];

        if (i >= 3) {
            dp[i] += dp[i - 3];
        }
    }

    printf("%lld", dp[n]);

    free(dp);
    return 0;
}
```

	Input	Expected	Got	
✔	6	6	6	✔
✔	25	8641	8641	✔
✔	100	24382819596721629	24382819596721629	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

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